[NEW RESULTS]-KDD-1060

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1 Additional Experiment Results to Reviewer gq6s

2. Interpretability Validation (W1, W5)

To rigorously evaluate interpretability on real-world data, we track the `I(20)` and `I(100)` metrics, which represent strict and full-budget feature retention thresholds. Higher scores and lower average ranks indicate that the highlighted temporal evidence effectively preserves predictive signals.

Table 1 summarizes these faithfulness results under a fixed masking and retraining protocol. The central objective here is not to claim complete semantic interpretation, but rather to verify whether the top-ranked temporal regions retain actual predictive power across complex domains. Under the stringent `I(20)` threshold, AnchorMoE achieves 0.6419, exceeding both the strongest time-series specific baseline (FIT, 0.5760) and the strongest generic baseline (IG, 0.6189). Under `I(100)`, AnchorMoE reaches 0.8620, and its gradient-free variant reaches 0.8667, both outperforming Dynamask (0.8284). These results confirm that the temporal evidence identified by our model remains highly predictive and compact on real data.

Figure 1 provides a representative NATOPS case study to complement the quantitative results. It demonstrates that the empirical faithfulness gains are naturally accompanied by a compact and localized temporal focus, avoiding diffuse attribution across the sequence.

2 Additional Experiment Results to Reviewer Hjgq

1. Principal Explanation (W1, W3, W4, W5)

To clarify the function of gradients within our framework, Table 2 isolates their role during training. Under the real-data faithfulness protocol, the gradient-free variant of AnchorMoE still exceeds the Integrated Gradients baseline on the mean `I(100)` score (0.8667 vs. 0.8384). The standard AnchorMoE model, however, maintains an advantage under the more restrictive `I(20)` threshold. This performance pattern aligns with our design motivation: gradient magnitude acts as an effective instance-level calibration signal during the training phase, rather than serving as the final extracted explanation at inference.

Table 2: Gradient-role control analysis evaluating temporal faithfulness. Higher is better for `I(20)` and `I(100)`; lower is better for rank.

Method	I(20)	Rank	I(100)	Rank
AnchorMoE	**0.6419**	**2.57**	**0.8620**	**2.29**
AnchorMoE-gradient-free	**0.6338**	3.14	**0.8667**	**2.43**
IG	0.6189	**2.71**	0.8384	3.14

2. Internal Mechanism (W2, W7, W8, W9)

To address concerns regarding potential expert collapse, we comprehensively tracked the internal utilization statistics. Active expert rate defines the fraction of available experts utilized, while effective experts measure the diversity-adjusted utilization. For the confidence gating mechanism, the κ mean, alongside specific thresholds for low ($\kappa < 0.1$) and high ($\kappa > 0.9$) activations, tracks the sparsity and selectivity of our decision binding.

Tables 3 through 6 systematically decouple three critical mechanism inquiries: whether expert utilization remains non-trivial, whether the routed content suffers from redundancy, and whether the confidence gate degenerates into uniform behavior. Table 3 reports the aggregate statistics across all 29 datasets. Expanding on this, Tables 4 and 5 demonstrate that the routing concentration and confidence sparsity dynamically adapt to dataset characteristics rather than collapsing into a single global pattern.

Crucially, Table 6 examines the structural necessity of our routing mechanism. We observe that removing DPAR drastically increases the inter-anchor redundancy from 0.3488 to 0.7213, resulting in a severe 106.8% surge. This spike confirms that DPAR functions primarily as a redundancy-reducing geometric prior enforcing diverse specialization, rather than a simplistic traffic-balancing mechanism.

Table 7 further breaks down targeted internal design choices. Removing the sample-level relevance cue causes the mean accuracy to degrade significantly from 0.9686 to 0.7084, whereas ablating the spectral cue maintains the block average. This behavior accurately reflects our architectural intent: the relevance cue is primarily responsible for injecting sequence-level context into the routing decisions, while the spectral branch functions as a stabilizing mechanism for periodic signals rather than the primary discriminative driver. Detailed sensitivity analysis regarding the hyperparameters is provided subsequently in Section 4.

3 Additional Experiment Results to Reviewer gMhY

1. Scalability Analysis & Runtime (W1)

To evaluate computational scalability, we report throughput in terms of inference-plus-explanation iterations per second, effectively charging the post-hoc explanation overhead to the black-box baselines. Peak memory tracks the maximum runtime footprint in MB.

Table 8 and Figure 3 present performance metrics scaled up to extreme 10k configurations. Rather than merely comparing raw forward propagation speeds, we focus on the end-to-end throughput required for transparent decision-making. Under this comprehensive evaluation, AnchorMoE achieves 122.8 iterations per second on the 10k feature dimension setting, compared to 30.4 for DLinear. Similarly, on the 10k sequence length setting, it reaches 57.1 iterations per second against 6.7 for TSLANet. These benchmarks demonstrate a concrete operational efficiency advantage in large-scale deployments, complementing our theoretical linear complexity.

Table 1: Real-data faithfulness evaluation. The top block reports summary statistics across datasets, while the lower block provides specific dataset-level measurements. Higher is better for `I(20)` and `I(100)`; lower is better for rank.

Dataset / Stat.	Metric	AnchorMoE	AnchorMoE-gradient-free	IG	DeepLift	Grad-CAM	Random	Dynamask	FIT
Summary statistics									
Overall mean	I(20)	**0.6419**	**0.6338**	0.6189	0.6168	0.5891	0.5301	0.5390	0.5760
Overall rank	I(20)	**2.86**	3.43	**3.14**	3.86	4.71	6.29	6.00	5.71
Overall mean	I(100)	**0.8620**	**0.8667**	0.8384	0.8363	0.8278	0.8035	0.8284	0.8264
Overall rank	I(100)	**2.57**	**2.86**	3.71	4.43	5.86	6.29	4.57	5.71
Dataset-level values									
ArticularyWordRecognition	I(20)	0.6554	0.6625	**0.6808**	**0.6749**	0.5832	**0.6749**	0.5757	0.6361
ArticularyWordRecognition	I(100)	0.8772	0.8810	0.8812	**0.8963**	0.8599	**0.8938**	0.8812	0.8725
BasicMotions	I(20)	**0.5750**	**0.5938**	0.4625	0.5094	0.5521	0.4583	0.5083	0.4531
BasicMotions	I(100)	**0.8175**	**0.8675**	0.7354	0.7306	0.7792	0.7388	0.8092	0.7819
Cricket	I(20)	0.6597	0.6111	**0.6985**	**0.6997**	0.6140	0.6198	0.6024	0.6869
Cricket	I(100)	0.8993	0.8819	**0.9041**	**0.9078**	0.8703	0.8443	0.8760	0.8892
FordA	I(20)	0.6661	**0.6821**	0.6301	0.6412	0.6469	0.4620	**0.6920**	0.5631
FordA	I(100)	0.8678	**0.8766**	0.8401	0.8287	0.8541	0.7505	**0.8800**	0.8061
NATOPS	I(20)	0.6764	0.6708	**0.6792**	**0.6769**	0.6069	0.5315	0.4843	0.5486
NATOPS	I(100)	**0.8883**	**0.8808**	0.8744	0.8671	0.8517	0.8500	0.7591	0.8252
RacketSports	I(20)	**0.5576**	0.5206	0.4729	0.4482	**0.5414**	0.4238	0.4833	0.4512
RacketSports	I(100)	**0.7566**	**0.7541**	0.7054	0.7053	0.7194	0.6762	0.7278	0.6910
SpokenArabicDigits	I(20)	**0.7030**	0.6959	**0.7081**	0.6675	0.5790	0.5405	0.4271	0.6928
SpokenArabicDigits	I(100)	**0.9274**	0.9251	**0.9283**	0.9180	0.8600	0.8711	0.8658	0.9191

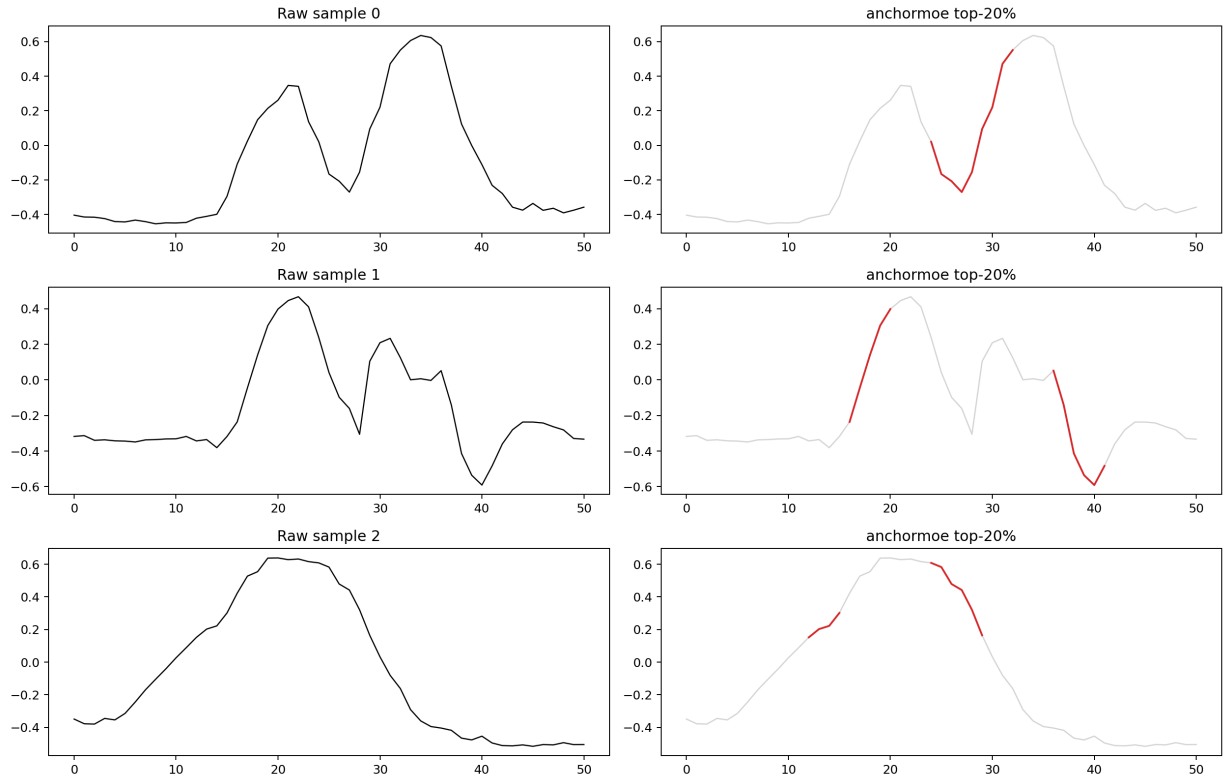


Figure 1: Representative NATOPS case study. Temporal evidence highlighted by AnchorMoE remains localized and compact.

Table 3: Aggregate mechanism statistics across the 29-dataset diagnostic sweep, tracking utilization and gating sparsity.

Metric	Mean	Median	Min	Max
active_expert_rate	0.5388	0.5000	0.1667	1.0000
max_expert_share	0.7674	0.7578	0.4705	1.0000
effective_experts	1.8952	1.9053	1.0000	3.2933
κ_mean	0.1844	0.1662	0.0001	0.6128
κ_low_ratio	0.5653	0.6325	0.0000	1.0000
κ_high_ratio	0.0272	0.0002	0.0000	0.2603

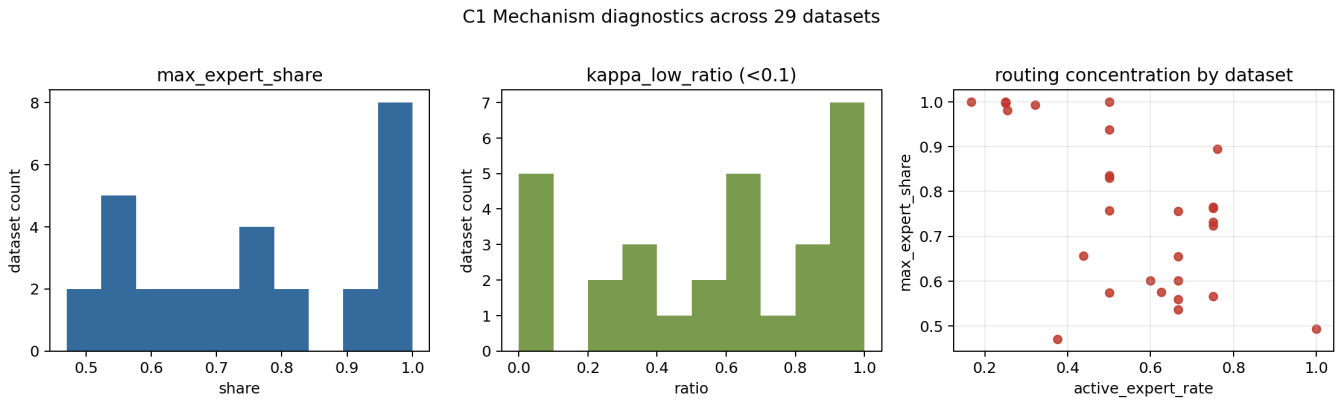


Figure 2: Mechanism diagnostics across 29 datasets, highlighting dataset-dependent utilization and confidence behavior.

Table 4: Dataset-specific expert utilization behavior. The model naturally adapts to varying routing regimes.

Dataset	Active rate	Max share	Effective experts
Libras	1.0000	0.4932	3.2933
Cricket	0.6667	0.5594	3.0690
PhonemeSpectra	0.3750	0.4705	2.9461
BasicMotions	0.7500	0.5656	2.2905
Heartbeat	0.2500	1.0000	1.0003
MotorImagery	0.1667	1.0000	1.0000

Table 5: Dataset-specific κ statistics, highlighting how the gating mechanism adjusts its sparsity depending on the data complexity.

Dataset	κ mean	Low ratio	High ratio
Libras	0.61276	0.07514	0.22591
Handwriting	0.44180	0.33335	0.26026
NATOPS	0.11251	0.81994	0.03575
SpokenArabicDigits	0.08893	0.86469	0.03999
Heartbeat	0.01114	0.97854	0.00000
MotorImagery	0.00007	0.99996	0.00000

Table 8: Throughput and peak memory scaling evaluated at 10k limits. Throughput accounts for both inference and explanation derivation.

Sweep	Point	AnchorMoE	Peak MB	Best baseline	Baseline	Rank
feature_dim	10000	122.822	535.1	DLinear	30.430	1
seq_len	10000	57.059	230.3	TSLANet	6.659	1

2. Interpretability on Real Data (W2)

As discussed in Section 1, we employ the same strict real-data faithfulness protocol utilizing `I(20)` and `I(100)` to measure predictive temporal localization. Please refer back to Table 1 and Figure 1 for detailed comparisons proving that our framework consistently preserves structural integrity on real-world datasets.

4 Additional Experiment Results to Reviewer ncgk

1. Tuning Complexity & Anchor Determination (W1, W3)

To assess tuning sensitivity, Table 9 records parameter responses across anchor diversity, spectral weight, group count, and confidence regularization. The empirical evidence counters the assumption of uniform brittleness. While factors such as anchor diversity and spectral weight present relatively stable operating regions, the confidence regularization weight requires careful and conservative tuning. Notably, aggressive penalization on the complex NATOPS dataset causes accuracy to collapse abruptly to 0.1667.

Consequently, we suggest a reproducible plateau-based heuristic for determining the optimal number of groups: selecting the minimal count that achieves validation saturation. Table 10 supports this approach, showing that the dataset-specific optimum consistently falls within a moderate range of 3 to 5 groups.

Table 9: Sensitivity analysis summary detailing operating ranges for key structural hyperparameters.

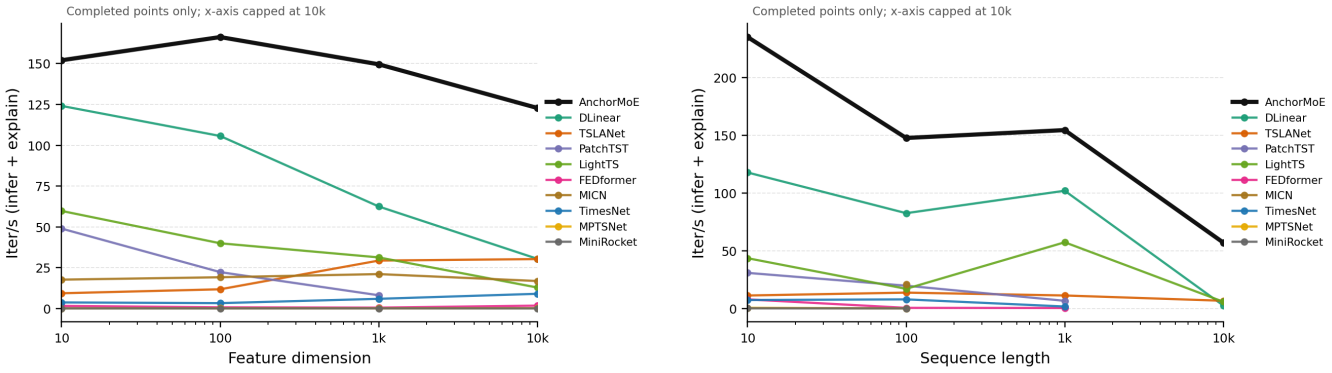
Parameter	Scanned values	Acc. range	Reading
anchor div.	0, 0.02, 0.05, 0.08, 0.1	0.9268–0.9332	Broadly workable operating range.
spectral weight	0, 0.1, 0.3, 0.5, 0.7	0.9281–0.9374	Moderate values maintain aggregate stability.
num groups	2, 3, 4, 5, 6, 7, 8	0.9166–0.9722	Plateau heuristic effectively yields moderate counts.
conf lambda	0, 0.005, 0.01, 0.02, 0.03	0.7976–0.9314	Highly sensitive; risks severe performance drops if over-penalized.

Table 6: B2 routing diagnostics. Lower redundancy indicates better structural diversity among the expert assignments.

Variant	Active rate	Max share	Effective experts	κ mean	Low ratio	High ratio	Redundancy
Full	0.4389	0.8525	1.4280	0.5319	0.1483	0.2606	0.3488
NoDPAR	0.5470	0.8283	1.4563	0.5600	0.1432	0.2865	0.7213
WeakDPAR	0.4278	0.8432	1.4518	0.5306	0.1528	0.2714	0.3788

Table 7: Targeted internal-design ablations. Experiments are grouped into discrete blocks (B1: MVSE, B2: DPAR, B3: UADB) evaluated on their respective diagnostic subsets. Because these subsets vary, absolute scores differ across blocks and are not directly comparable to the 18-dataset overall averages in the main paper. Higher is better.

Block	Variant	Mean Acc	Mean F1	Mean I(20)	Mean I(100)
B1	Full	0.9686	0.9685	0.9250	0.9348
B1	NoRelevance	0.7084	0.6687	0.8478	0.9348
B1	NoSpectral	0.9737	0.9736	0.9268	0.9348
B2	Full	0.9686	0.9685	0.9250	0.9348
B2	NoDPAR	0.9727	0.9726	0.9580	0.9348
B2	WeakDPAR	0.9734	0.9732	0.9518	0.9348
B3	Full	0.9509	0.9504	0.8953	0.9081
B3	AnchorMoE-gradient-free	0.9638	0.9638	0.9097	0.9051
B3	NoConfidence	0.9477	0.9475	0.8819	0.9078



(a) Feature-dimension sweep. **(b) Sequence-length sweep.**

Figure 3: End-to-end throughput trends evaluated across increasingly complex dimensions up to the 10k scale.

Table 10: Dataset-specific optimal group counts, confirming that performance consistently saturates within a moderate range.

Dataset	Best num_groups	Acc	F1	AOPCAcc	IoU
ArticularyWord Recognition	3	0.9733	0.9733	0.7898	0.2820
Cricket	5	0.9722	0.9720	0.7000	0.2796
FordA	3	0.9356	0.9356	0.3461	0.2499
Heartbeat	4	0.7707	0.7674	0.4898	0.2769
NATOPS	4	0.9833	0.9833	0.6578	0.3750
SpokenArabic Digits	3	0.9900	0.9900	0.7773	0.3007

2. Risk of Over-filtering & Safety-Critical Trade-offs (W2)

Building on the real-data evaluations in Table 1, Table 11 investigates the risk of over-filtering by ablating the intrinsic confidence score entirely. Without this self-assessment capability, the model reverts to a uniform gating regime where κ defaults to 1.0, treating every routed patch identically regardless of its actual informativeness. As background noise dilutes the genuine discriminative signals, the strict early-stage feature localization score, `I(20)`, naturally drops from 0.8953 to 0.8819. This structural degradation validates our core design premise: while the routing mechanism governs spatial feature allocation, the confidence score provides an indispensable filtering layer to suppress weak evidence. These diagnostics jointly map out a safe and effective operating boundary for deploying the framework.

Table 11: Selective-binding diagnostics. Ablating confidence weighting forces uniform signal admission and degrades early-stage feature localization.

Variant	I(20)	I(100)	Mean κ	$\kappa < 0.1$	$\kappa > 0.9$
Full	0.8953	0.9081	0.3833	0.0989	0.0504
AnchorMoE-gradient-free	0.9097	0.9051	0.2322	0.4188	0.0049
NoConfidence	0.8819	0.9078	1.0000	0.0000	1.0000

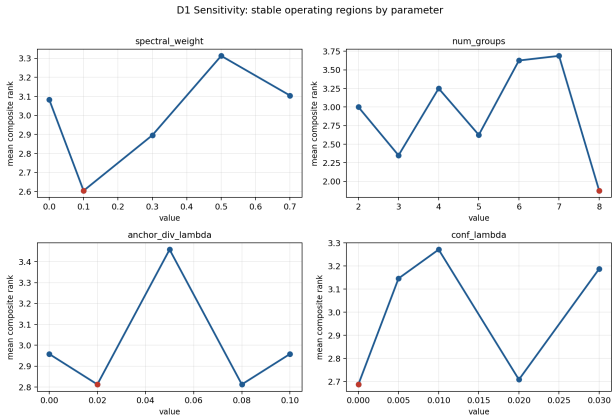


Figure 4: Stable operating regions for primary hyperparameters.